

$$1) A = [-1; 1; 0]$$

$$V = \frac{|ax+by+cz+d|}{\sqrt{a^2+b^2+c^2}}$$

$$\mathcal{P}: x = 1 - 2t - s$$

$$y = 2 - s \Rightarrow s = 2 - y$$

$$z = 1 - t \Rightarrow t = 1 - z$$

$$x = 1 - 2 + 2z - 2 + y$$

$$0 = x - y - 2z + 3$$

$$V = \frac{|-1-1+3|}{\sqrt{1+1+4}} = \frac{1}{\sqrt{6}}$$

$$2) A = [-1; 1; 0]$$

$$\mathcal{P}: x = 1 - 2t$$

$$y = 2 - t$$

$$z = 1 + t$$

$$\vec{S}_P = (-2; -1; 1) = \vec{n}_P$$

$$\mathcal{P}: -2x - y + z + d = 0$$

$$A \in \mathcal{P}: 2 - 1 + d = 0$$

$$d = -1$$

$$\mathcal{P}: -2x - y + z - 1 = 0$$

$$-2x - y + z - 1 = 0$$

$$-2 + 4t - 2 + t + t - 1 = 0$$

$$6t - 4 = 0$$

$$6t = 4$$

$$t = \frac{4}{6}$$

$$t = \frac{2}{3}$$

$$x = 1 - 2 \cdot \frac{2}{3} = -\frac{1}{3}$$

$$y = 2 - \frac{2}{3} = \frac{4}{3}$$

$$z = 1 + \frac{2}{3} = \frac{5}{3}$$

$$P = [-\frac{1}{3}; \frac{4}{3}; \frac{5}{3}]$$

$$V = |\vec{n}P| = \sqrt{(-\frac{1}{3} + 1)^2 + (\frac{4}{3} - 1)^2 + (\frac{5}{3})^2} = \frac{\sqrt{30}}{3}$$